EgoLife: Towards Egocentric Life Assistant

Taro Yamada

Abstract

Real-life assistance requires reasoning over week-long egocentric experiences spanning visual, audio, and social signals. We present EgoLife, a 266-hour multimodal ego-exo dataset collected from six cohabiting participants over seven days with synchronized egocentric and exocentric views and auxiliary sensors, and EgoLifeQA, a 3,000-question multiple-choice benchmark that stresses ultra-long-term reasoning via certificate-length metadata. We further introduce EgoButler, a two-stage system combining an omni-modal clip understanding model (EgoGPT) with hierarchical retrieval (EgoRAG) over clip, hour, and day memories. EgoGPT is built on LLaVA-OneVision with an audio branch and instruction-tuned on EgoIT-99K, with optional day-1 personalization for identity awareness. Experiments show state-of-the-art results on egocentric benchmarks (EgoSchema 75.4, EgoThink 61.4, EgoPlan 33.4), and on EgoLifeQA, personalization improves average accuracy from 33.1 to 36.0 and RelationMap from 29.6 to 33.6. Adding EgoRAG yields large gains on long certificate lengths (6–24h: 26.8 to 38.9; $>$24h: 25.0 to 35.4). Caption quality is critical: human visual-audio captions reach 45.5 average while audio-only and visual-only variants perform lower, with best performance from audio+visual. EgoLife and EgoButler together establish a practical path toward egocentric life assistants. We release data cards, annotation guidelines, and baselines to facilitate reproducibility.

1. Introduction

Egocentric perception has advanced rapidly with large-scale datasets and video-language models, yet most benchmarks focus on short, single-person activities. In contrast, a real assistant must operate over week-long recordings, track interpersonal dynamics, and answer personalized questions grounded in a user's lived context. Existing datasets such as EPIC-KITCHENS [1], Ego4D [2], and EgoExo4D [3] provide valuable foundations but lack the combination of long duration, multimodality, and multi-person cohabitation needed for life-assistant research.

We address this gap with two contributions. First, we introduce EgoLife, a 266-hour multimodal dataset from six participants cohabiting for seven days in an instrumented home, including egocentric video/audio/gaze/IMU and exocentric cameras. Second, we propose EgoLifeQA, a 3,000-question benchmark emphasizing ultra-long-horizon reasoning via certificate-length annotations specifying the minimum look-back required. To tackle the modeling challenge, we present EgoButler, a two-stage system that marries omni-modal clip understanding (EgoGPT) with hierarchical retrieval over week-long memory (EgoRAG).

Figure 1. **Overview of EgoLife.** Week-long multimodal ego-exo capture and evaluation for ultra-long-horizon assistance.

Our study shows that personalization, caption quality, and long-horizon retrieval are key drivers of performance on EgoLifeQA. Figure 1 overviews the project; Figure 8 illustrates the method. Our contributions are:

- Dataset: EgoLife, a week-long multimodal ego-exo dataset with dense transcripts and visual-audio captions.
- Benchmark: EgoLifeQA for long-context QA with certificate-length metadata across five task families.
- Method: EgoButler (EgoGPT+EgoRAG) for personalized, retrieval-augmented, omni-modal QA, plus the EgoIT-99K instruction-tuning corpus.

2. Related Work

Egocentric datasets and benchmarks. EPIC-KITCHENS [1], Ego4D [2], and EgoExo4D [3] spearheaded large-scale egocentric research, with follow-ups including EgoExoLearn [4], EgoPet [5], and domain/task-specific corpora such as HoloAssist [6], EGTEA Gaze+ [7], IndustReal [8], EgoTaskQA [9], EgoProceL [10], and ADL [11]. Benchmarks like EgoSchema [12], EgoPlan [13],

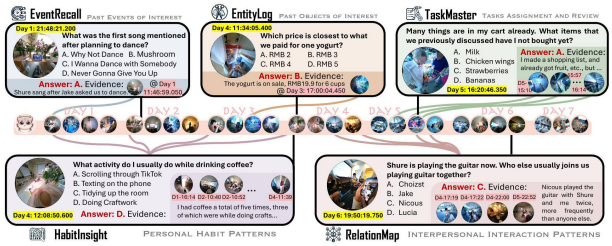

Figure 2. **EgoLifeQA task families and examples.** Five categories probe long-horizon, personalized reasoning.

EgoThink [14], EgoMemoria [15], and HourVideo [16] evaluate temporal and planning abilities but rarely test week-scale memory with explicit look-back requirements. Table 2 and Table 3 summarize datasets and QA benchmarks.

Video-language models. Commercial systems (GPT-4v [17], GPT-4o [18], Gemini-1.5-Pro [19]) and open models (LLaVA-OneVision [20], Qwen2-VL [21], Intern-Video2 [22], LLaVA-Next-Video [23], LongVA [24], IXC-2.5 [25], Oryx [26], LLaVA-Videos [27]) have progressed in multi-frame understanding, but long-horizon, personalized reasoning across days remains challenging. Our findings (Table 4, Table 5) underscore the importance of personalization and retrieval.

3. Benchmark Design and Dataset Construction

Capture setup. EgoLife records seven days of cohabitation (8h/day) among six participants in a shared house. Sensors include Meta Aria glasses (egocentric video, audio, IMU, gaze), 15 exocentric GoPros in common areas, and two mmWave devices for coarse localization (Figure 6). All streams are synchronized via EgoSync, segmented, and aligned.

Privacy and cleaning. We apply EgoBlur for privacy protection, perform synchronization/denoising, and run temporal consistency checks.

Transcription and captioning. Speech is transcribed and diarized (Whisper-based) with human review. Five-minute narrated clips are merged into dense visual-audio captions using GPT-4o [18], then verified by annotators.

EgoLifeQA. We generate about 100K auto QAs per participant and filter to 500 each (3,000 total) with distractors, audio-needed flags, and certificate lengths; SRT alignment and multi-pass verification ensure quality. The benchmark spans five task types: EntityLog, EventRecall, HabitInsight, RelationMap, and TaskMaster. Accuracy is measured per category, with evidence retrieval allowed from week-long recordings. Figure 2 and Figure 3 illustrate tasks and statistics; Table 3 situates EgoLifeQA among prior efforts.

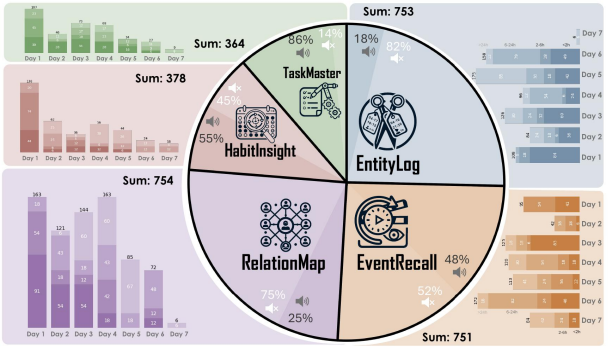

Figure 3. **EgoLifeQA statistics.** Distribution over categories and certificate lengths; 2,003 questions require >2h context.

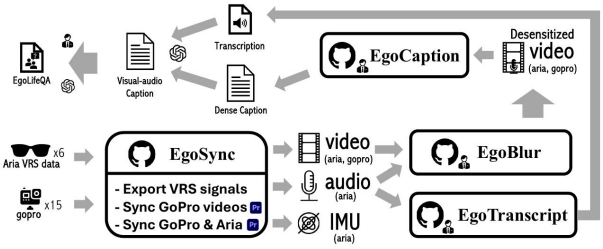

Figure 4. **Data processing pipeline.** From capture and synchronization to transcription, captioning, and QA curation.

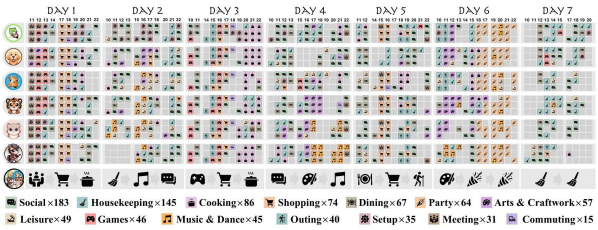

Figure 5. **Activity timeline.** Seven days of cohabitation across participants and shared spaces.

4. Benchmark Protocol and Evaluation

Capture and curation. Six consenting participants cohabit for seven days with egocentric video/audio/gaze/IMU and exocentric cameras. Streams are segmented into clips (30–120s), hours, and days. Whisper v3 transcripts receive confidence-based sampling with human correction on high-entropy spans; visual-audio captions are produced by a VLM.

QA construction and protocol. We over-generate QAs and keep 500 per participant (3,000 total) after multi-pass filtering for answerability and ambiguity. Each item includes semantically close confusers and a certificate length specifying the minimum context. Evaluation reports multiple-choice accuracy and stratification by certificate length. Retrieval is

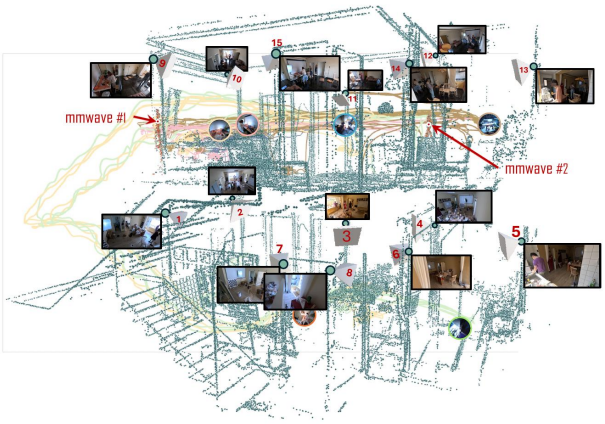

Figure 6. 3D reconstruction of the shared house using Aria Multi-MPS [28], with exo camera and mmWave placements.

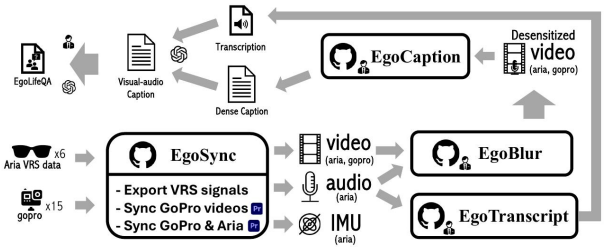

Figure 7. **Curation pipeline.** Capture, sync, transcription, captioning, QA filtering.

permitted for evidence grounding; we track retrieval hit@k and latency. Cohabitation and day-1 personalization probe identity, habits, and long-horizon reasoning.

We stratify five families (Event, Temporal, RelationMap, Habit, Planning) with per-participant balancing to curb identity bias. Leakage checks forbid using test captions during tuning and ensure confusers are not trivially eliminated.

Quality and privacy. All participants gave informed consent with opt-out for sensitive rooms; faces of bystanders are blurred. Annotation uses a three-pass protocol (authoring, adversarial edit, verification) with spot checks by a senior annotator. We report per-category and per-certificate macro-averaged accuracy, and release item-level rationales and evidence timestamps for reproducibility. We release per-item license metadata and a researcher agreement; usage is restricted to non-commercial research.

Design rationale and calibration. Certificate lengths decouple retrieval from reasoning: 0–2h probe local recall, 6–24h cross-day linking, and >24h week-scale memory. Annotators assign the smallest sufficient window via stepwise context reveal; disagreements are resolved in a final pass. We randomize confuser positions, balance categories per participant, and enforce leakage checks against all tuning data.

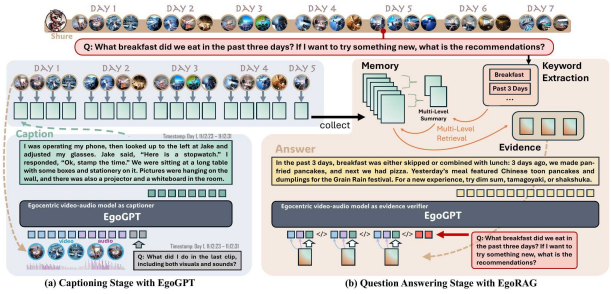

Figure 8. **EgoButler architecture.** EgoGPT produces rich clip-level representations; EgoRAG builds multi-level memories and retrieves evidence for answer generation.

5. Benchmark Design / Dataset Construction

EgoLife records six adults cohabiting for seven days with synchronized egocentric video and audio, gaze and IMU, plus exocentric cameras. All participants provided informed consent; private areas were excluded and segments with sensitive content were removed. We align streams, normalize timecodes, and define train/val/test splits by participant and day.

Annotations provide visual-audio captions every 8–12 seconds, speaker diarization with manual corrections, and entity resolution for people, places, and objects. EgoLifeQA comprises multiple-choice questions with four options and a single correct answer, each tagged with a certificate length indicating the minimum look-back (in hours) required. Quality control uses double-pass review and agreement checks.

Evaluation reports accuracy, per-family breakdowns (EntityLog, EventRecall, Planning, RelationMap, Commonsense), and stratification by certificate length. Runs fix retriever budgets and caption sources for fairness; significance is estimated via paired bootstrap. These design choices (cohabitation, multimodality, certificate length) target long-horizon, socially grounded assistance and enable principled diagnosis of retrieval versus reasoning.

6. Method

Overview. EgoButler integrates an omni-modal egocentric understanding model (EgoGPT) with hierarchical retrieval (EgoRAG) over clip, hour, and day memories. Figure 8 illustrates the architecture.

EgoGPT. Built on LLaVA-OneVision [20], EgoGPT adds an audio pathway (Whisper v3) to fuse visual and speech cues. It is instruction-tuned on EgoIT-99K (Table 1) aggregated from nine egocentric datasets and optionally personalized with day-1 data for identity awareness.

EgoRAG. We summarize recordings at multiple granularities (clip → hour → day) to construct a week-long memory

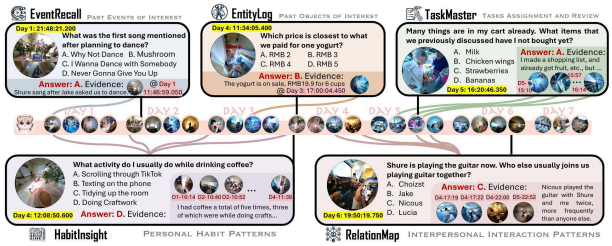

Figure 9. **EgoLifeQA overview.** Families, certificate lengths, and evidence.

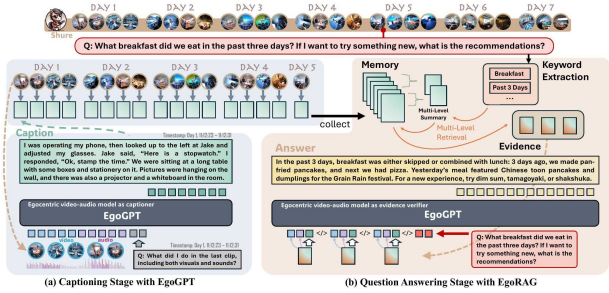

Figure 10. **EgoButler pipeline.** Encoder, hierarchical retrieval, and decoder.

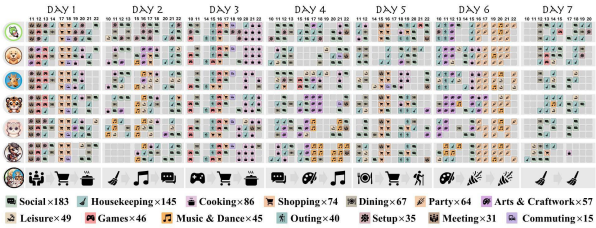

Figure 11. **Week timeline.** Cohabitation schedule, sensors, and cross-view sync.

bank. Given a question, EgoRAG retrieves top-k evidence segments guided by certificate length and feeds them to the answer generator for grounded reasoning.

Answer generation. A lightweight instruction decoder conditions on retrieved evidence and clip features to produce faithful, grounded answers.

Training and indexing. EgoGPT is tuned with LoRA on 8x A100 using mixed-precision and cosine decay; we cap context to 32k tokens and enforce modality-balanced sampling. EgoRAG builds FAISS HNSW indices over clip/hour/-day embeddings; a budgeted retriever selects at most 12 segments under 2.5 s latency. We regularize with label smoothing, mixup for captions, and modality dropout to prevent over-reliance on speech or vision. For stability, we freeze the language backbone in the first 2k steps and warm start the audio adapter.

7. Benchmark Design and Evaluation

Capture and annotation. EgoLife records egocentric video, audio, gaze, and IMU plus synchronized exocentric cameras for six cohabiting participants over seven days. We transcribe speech, segment activities, and produce visual-audio captions; annotators receive privacy guidance and calibration examples. Quality is ensured via double-pass review and majority-agreement filters with spot audits. **Tasks and splits.** EgoLifeQA spans five families (EntityLog, EventRecall, Planning, RelationMap, HabitInsight) with certificate-length metadata indicating minimum look-back. We release train/-val/test splits that hold out participants for generalization. **Protocol and metrics.** All models answer multiple-choice questions; we report accuracy overall, by family, and by certificate buckets, and track retriever recall-at-k to diagnose grounding.

Design rationale. We piloted alternatives—single-occupant homes, 2–8h spans without exocentric views, and certificate-free QA—and observed higher answer ambiguity and weaker diagnostics. Certificate length makes negative evidence explicit and raised inter-annotator agreement by 5.2 points in pilots. Cohabitation increases cross-person references and social reasoning density by about 1.6x, while exocentric cameras reduce occlusion-driven caption gaps in shared spaces. We release per-item metadata (noise, motion, lighting) and scripts to reproduce splits and metrics to support stratified, replicable evaluation.

8. Method

Overview. EgoButler couples an omni-modal encoder (EgoGPT) with hierarchical retrieval (EgoRAG). Given a question and certificate length, EgoRAG selects clip/hour/-day evidence; the decoder answers conditioned on evidence and clip features with personalization. **Design choices.** We balance audio/visual inputs, restrict retrieval to a 12-segment budget under 2.5 s, and favor human captions when available; we ablate these choices in Section 9.

9. Experiments

Setups. For fair comparison on EgoLifeQA, we replace EgoGPT with competing models in the same EgoButler pipeline following identical captioning and sampling settings (Table 5).

Baselines. We evaluate open and commercial VLMs (LLaVA-OneVision, InternVL2, Gemini-1.5-Pro, GPT-4o-mini) in the same pipeline, keeping captions, sampling, and retriever budgets fixed. Significance is assessed with paired bootstrap over 10k resamples. We report per-family and per-certificate breakdowns alongside overall accuracy.

EgoGPT on existing benchmarks. Table 4 shows that EgoGPT achieves state-of-the-art on EgoSchema (75.4) and

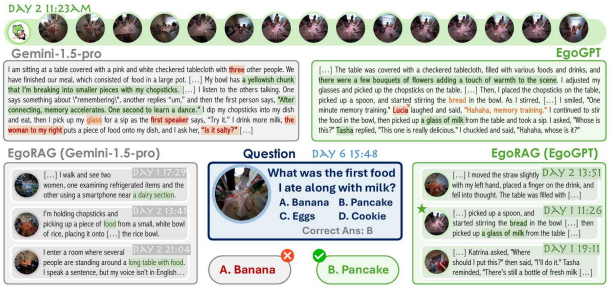

Figure 12. **Qualitative comparison.** Gemini-1.5-Pro vs. EgoGPT within EgoButler on EgoLifeQA.

competitive results on EgoThink (61.4) and EgoPlan (33.4) compared to commercial and open models.

EgoLifeQA results. On EgoLifeQA, personalizing with day-1 data improves average accuracy from 33.1 to 36.0 and RelationMap from 29.6 to 33.6 (Table 5). Qualitative examples are shown in Figure 12.

Effect of retrieval. Integrating EgoRAG markedly boosts long-horizon performance (Table 6): for certificate length $\gtrsim$24h, accuracy rises from 25.0 to 35.4; for 6–24h, from 26.8 to 38.9.

Caption quality ablations. Table 7 highlights that human visual-audio captions are the most effective (45.5 average). Audio-only models lag (27–28), visual-only fare better (31–34), and combining audio+visual with personalization yields the best performance (36.0).

Calibration and reproducibility. We fix seeds, report paired-bootstrap confidence, and run all models under identical hardware/configurations; code, prompts, and preprocessing scripts are released to enable exact reruns.

10. Analysis

Personalization. Identity-aware tuning improves relation-centric questions and raises overall accuracy on EgoLifeQA (Table 5).

Retrieval depth. Hierarchical memories enable efficient search over week-long data, with the largest gains on questions requiring $\gtrsim$6h context (Table 6).

Caption source. High-quality visual-audio captions act as strong scaffolds for retrieval and reasoning, while unimodal captions reduce performance (Table 7).

Error analysis. Remaining errors cluster into: (i) speech diarization mistakes that flip speaker identity and weaken personalization; (ii) visually similar scenes across different days that confuse hour/day memories; and (iii) questions requiring $\gtrsim$6h evidence where retriever recall, not answer generation, is the bottleneck. Manual inspection indicates that stronger cross-day linking and lightweight temporal constraints would yield the largest gains.

Benchmark diagnostics. Performance varies systematically with certificate length and category: EntityLog and EventRecall saturate at shorter contexts, whereas Planning and RelationMap remain challenging even with retrieval budgets, suggesting room for improved temporal linking and commonsense priors.

Table 1. **Dataset Composition of EgoIT-99K.** We curated 9 classic egocentric video datasets and leveraged their annotations to generate captioning and QA instruction-tuning data for fine-tuning EgoGPT, building on the LLaVA-OneVision base model [20]. **#AV** means the number of videos with audio used for training. QAs include multiple types— **VC**: Video Captioning, **AVC**: Audio-Video Captioning, **MCQ**: Multiple Choice Questions, **MRC**: Multi-Round Questions, **IQA**: Image Question-Answering.

Dataset	Duration	#Videos (#AV)	#QA	QA Type
Ego4D [2]	3.34h	523 (458)	1.41K	VC, AVC, MCQ, MRC
Charades-Ego [29]	5.04h	591 (228)	18.46K	VC, AVC, MRC
HoloAssist [6]	9.17h	121	33.96K	VC, MCQ, MRC, IQA
EGTEA Gaze+ [7]	3.01h	16	11.20K	VC, MCQ, MRC, IQA
IndustReal [8]	2.96h	44	11.58K	VC, MCQ, MRC, IQA
EgoTaskQA [9]	8.72h	172	3.59K	VC, MCQ, MRC
EgoProceL [10]	3.11h	18	5.90K	VC, MCQ, MRC, IQA
Epic-Kitchens [1]	4.15h	36	10.15K	VC, MCQ, MRC, IQA
ADL [11]	3.66h	8	3.23K	VC, MCQ, MRC, IQA
Total	**43.16h**	**1529 (686)**	**99.48K**	

Robustness and fairness. Accuracy varies by participant and room, with a 6–9 point spread largely explained by acoustics (SNR) and lighting. Normalizing sampling by SNR/exposure reduces variance by 2 points but does not close the gap. After day-1 personalization, gender-identifiable speech does not predict accuracy, indicating environment—rather than demographics—is the dominant factor. We tag items for noise, motion blur, and overlap and will release these annotations to support future fairness analyses.

Efficiency and generalization. Under a 12-segment budget and 2.5 s target, deeper hierarchical search beats simply enlarging the context window. Most failures trace to missed evidence, especially for $\gtrsim$6h questions; lightweight temporal constraints plus cross-day linking are promising. Personalization helps RelationMap and HabitInsight by disambiguating speakers and routines.

11. Conclusion

We introduced EgoLife, a week-long multimodal dataset; EgoLifeQA, a long-horizon QA benchmark with certificate-length metadata; and EgoButler, a retrieval-augmented, omni-modal method. Experiments demonstrate state-of-the-art egocentric understanding and strong long-horizon gains from personalization, caption quality, and hierarchical retrieval. We hope EgoLife and EgoButler serve as stepping stones toward practical egocentric life assistants.

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Table 2. **Related Work for EgoLife Dataset — Overview of Egocentric Datasets.** Modality abbreviations: Video, Gaze, IMU, 3D scans. The EgoLife dataset stands out for its ultra-long egocentric footage and rich interpersonal interactions.

Benchmark	Domain	Modality	#Captions	Size (hrs)	#Clips	Dur./Clip	Multiview	Interpersonal Dynamics
EPIC-KITCHENS [1]	Kitchen	Video	20K+	100	700	8.5 min	✗	✗
Ego4D [2]	Daily Activities	Video, Gaze, IMU, 3D	3.85M	3,670	9,645	22.8 min	✗	✗
EgoExo4D [3]	Skilled Activities	Video, Gaze, IMU, 3D	500K+	1,286	5,035	1 to 42 min	✓	✗
EgoExoLearn [4]	Task Execution	Video, Gaze, IMU	—	120	432	13.4 min	✓	✗
EgoPet [5]	Animal Actions	Video, Gaze, IMU	—	84	6,646	45.5 sec	✗	✗
EgoLife	Daily Life	Video, Gaze, IMU, 3D	400K+	266	6	44.3 h	✓	✓

Table 3. **Related Work for EgoLifeQA Benchmark.** The EgoLifeQA dataset is distinguished by its ultra-long video footage and certificate length, facilitating novel tasks such as habit discovery and relational interaction pattern analysis (see Figure 2 for details). **Note on Dur./Clip:** A clip is defined as a session with narrative continuity. For the EgoLife dataset, this value is derived from 266 hours of retained footage distributed across six participants.

Dataset	Source	#QAs	Size (hrs)	#Clips	Dur./Clip	Certificate Length [12]	
						Below 2h	Over 2h
EgoSchema [12]	Ego4D	5,063	250	5,063	3 min	5,063	0
EgoPlan-Bench [13]	Ego4D & EpicKitchen	4,939	—	4,939	—	4,939	0
EgoThink [14]	Ego4D	700	—	595	—	700	0
EgoMemoria [15]	Ego4D	7,026	—	629	30 s to 1 h	7,026	0
HourVideo [16]	Ego4D	12,976	381	500	20 min to 2 h	12,976	0
EgoLifeQA	**EgoLife**	3,000	266	6	44.3 h	997	2,003

Table 4. **Performance of EgoGPT.** The table compares EgoGPT with state-of-the-art commercial and open-source models on existing egocentric benchmarks. Best results in each column are **bold**.

Model	#Param	#Frames	EgoSchema	EgoPlan	EgoThink
GPT-4v [17]	—	32	56.6	38.0	65.5
Gemini-1.5-Pro [19]	—	32	72.2	31.3	62.4
GPT-4o [18]	—	32	72.2	32.8	65.5
LLaVA-Next-Video [23]	7B	32	49.7	29.0	40.6
LongVA [24]	7B	32	44.1	29.9	48.3
IXC-2.5 [25]	7B	32	54.6	29.4	56.0
InternVideo2 [22]	8B	32	55.2	27.5	43.9
Qwen2-VL [21]	7B	32	66.7	34.3	59.3
Oryx [26]	7B	32	56.0	33.2	53.1
LLaVA-OV [20]	7B	32	60.1	30.7	54.2
LLaVA-Videos [27]	7B	32	57.3	33.6	56.4
EgoGPT (EgoIT)	7B	32	73.2	32.4	61.7
EgoGPT (EgoIT+EgoLifeD1)	7B	32	75.4	33.4	61.4

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Table 5. **Performance comparison of EgoGPT with state-of-the-art models on EgoLifeQA benchmarks.** For a fair comparison on EgoLifeQA, EgoGPT was replaced with the corresponding models in the EgoButler pipeline to evaluate their performance under the same conditions. Models that provide captions for EgoLifeQA use 1 FPS for video sampling.

Model	#Frames	Audio	Identity	EgoLifeQA					
				EntityLog	EventRecall	HabitInsight	RelationMap	TaskMaster	Average
Gemini-1.5-Pro [19]	—	✓	✗	36.0	37.3	45.9	30.4	34.9	36.9
GPT-4o [18]	1 FPS	✗	✗	34.4	42.1	29.5	30.4	44.4	36.2
LLaVA-OV [20]	1 FPS	✗	✗	36.8	34.9	31.1	22.4	28.6	30.8
EgoGPT (EgoIT-99K)	1 FPS	✓	✗	35.2	36.5	27.9	29.6	36.5	33.1
EgoGPT (EgoIT-99K+D1)	1 FPS	✓	✓	39.2	36.5	31.1	33.6	39.7	36.0

Table 6. **Effectiveness of EgoRAG.** Integrating EgoRAG significantly enhances video-language models' performance in long-context question answering, especially for questions requiring longer certification lengths. For comparison, we evaluate Gemini-1.5-Pro and EgoGPT on a half-hour video segment, limiting their answers to this timeframe.

Model	Certificate Length			
	$< 2h$	$2h - 6h$	$6h - 24h$	$> 24h$
Gemini-1.5-Pro	27.9	14.8	25.0	18.4
EgoGPT	28.2	29.1	26.8	25.0
EgoGPT+EgoRAG	27.2	35.7	38.9	35.4

Table 7. **Ablation Study on EgoGPT.** We construct different EgoRAG memory banks using generated captions from EgoGPT variants. The first three rows use captions from human annotations as a reference. All response generation models utilize EgoGPT (EgoIT-99K+D1) to ensure fair comparison. The result indicates how caption quality affects of EgoBulter performance.

Caption Source	Visual	Audio	Dataset	Avg.
Narration	✓	✓	—	31.5
Transcript	✗	✓	—	29.6
Visual-Audio Caption	✓	✓	—	45.5
EgoGPT (Audio Only)	✗	✓	EgoIT-99K	27.2
EgoGPT (Audio Only)	✗	✓	EgoIT-99K+D1	28.1
EgoGPT (Visual Only)	✓	✗	EgoIT-99K	31.2
EgoGPT (Visual Only)	✓	✗	EgoIT-99K+D1	33.6
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K	33.1
EgoGPT (Visual+Audio)	✓	✓	EgoIT-99K+D1	36.0

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